

Digital Design

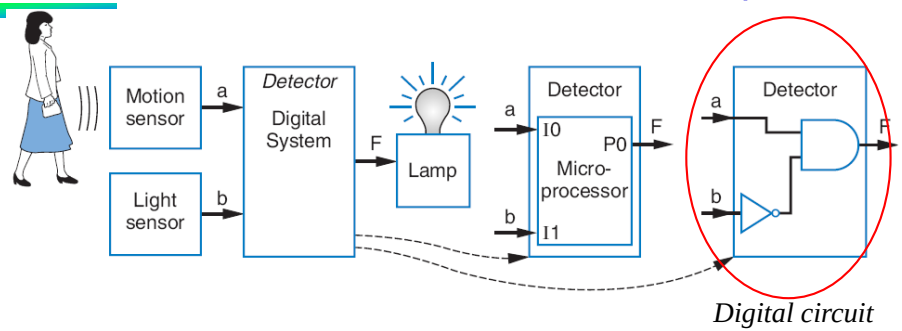
Chapter 2:
Combinational Logic Design

Slides to accompany the textbook Digital Design, with RTL Design, VHDL, and Verilog, 2nd Edition,
by Frank Vahid, John Wiley and Sons Publishers, 2010.
http://www.ddvahid.com

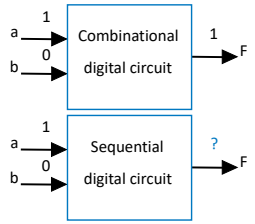
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Introduction — Combinational vs. Sequential



- Let's learn to design digital circuits
- We'll start with a simple form of circuit:
  - ***Combinational circuit***
    - A digital circuit whose outputs depend solely on the *present **combination** of the circuit inputs' values*

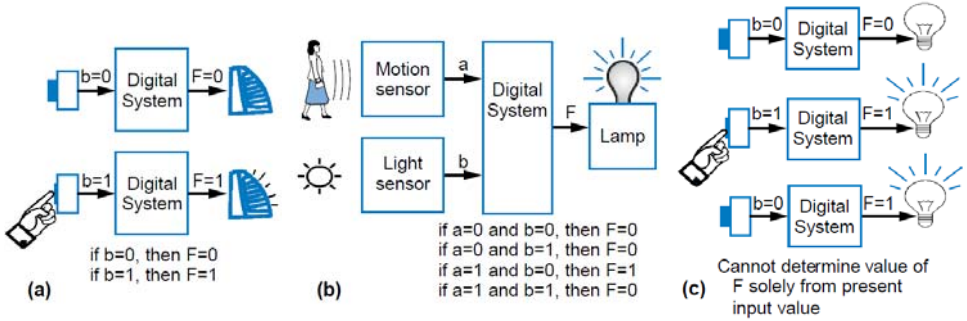




Note: Slides with animation are denoted with a small red "a" near the animated items

Combinational vs. Sequential

- A digital circuit whose outputs depend solely on the present combination of the circuit inputs' values is combinational Circuit.

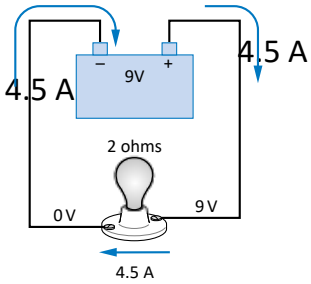


Combinational Design | **Combinational Design** | **Sequential Design**



Switches

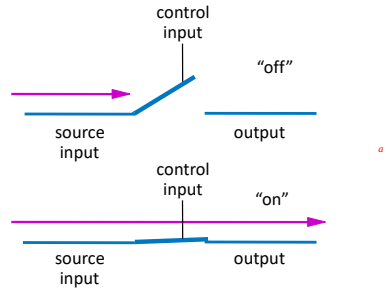
- Electronic switches are the basis of binary digital circuits
  - Electrical terminology
    - ***Voltage***: Difference in electric potential between two points
      - Analogous to water pressure
    - ***Current***: Flow of charged particles
      - Analogous to water flow
    - ***Resistance***: Tendency of wire to resist current flow
      - Analogous to water pipe diameter
    - V = I * R (Ohm's Law)

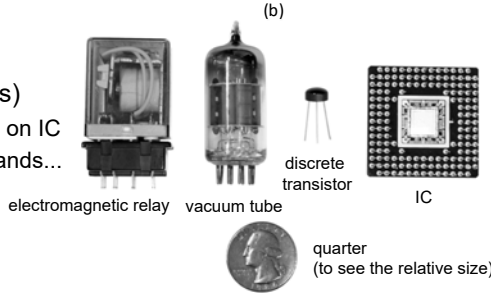




Switches

- A switch has three parts
  - Source input, and output
    - Current wants to flow from source input to output
  - Control input
    - Voltage that controls whether that current can flow
- The amazing shrinking switch
  - 1930s: Relays
  - 1940s: Vacuum tubes
  - 1950s: Discrete transistor
  - 1960s: Integrated circuits (ICs)
    - Initially just a few transistors on IC
    - Then tens, hundreds, thousands...



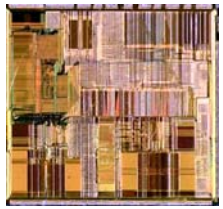


quarter (to see the relative size)

Moore's Law

- IC capacity doubling about every 18 months for several decades
  - Known as "Moore's Law" after Gordon Moore, co-founder of Intel
    - Predicted in 1965 predicted that components per IC would double roughly every year or so
  - Book cover depicts related phenomena
    - For a particular number of transistors, the IC shrinks by half every 18 months
      - Notice how much shrinking occurs in just about 10 years
      - Enables incredibly powerful computation in incredibly tiny devices
  - Today's ICs hold *billions* of transistors
    - The first Pentium processor (early 1990s) needed only 3 million

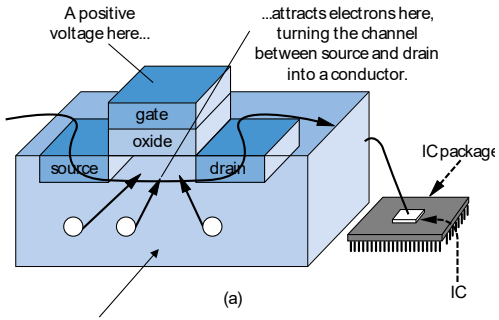




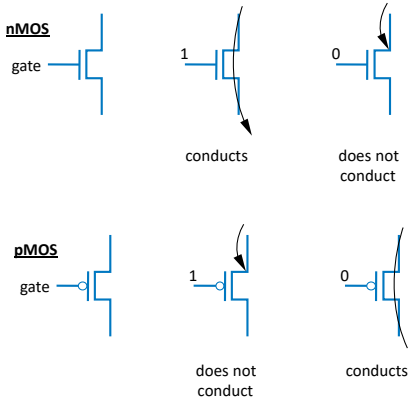
An Intel Pentium processor IC having millions of transistors

The CMOS Transistor

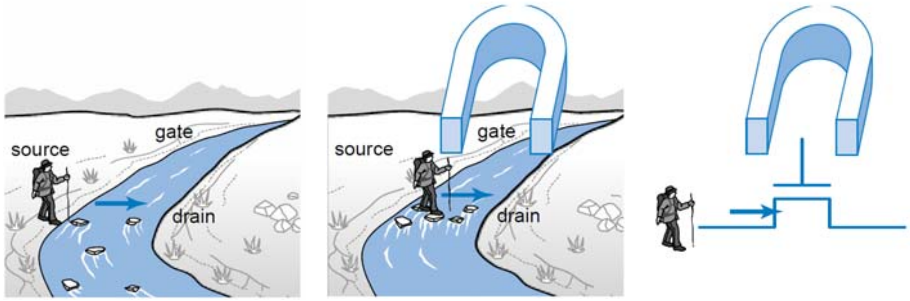
- CMOS transistor
  - Basic switch in modern ICs



Silicon -- not quite a conductor or insulator:
Semiconductor



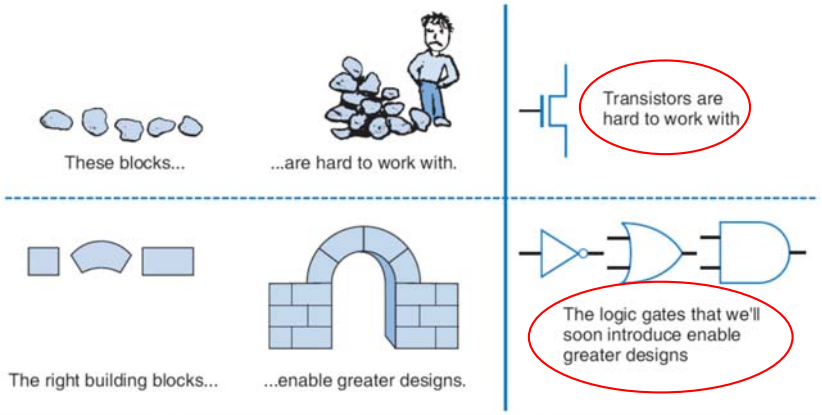
CMOS Transistor Analogy



Boolean Logic Gates

Building Blocks for Digital Circuits

(Because Switches are Hard to Work With)



- "Logic gates" are better digital circuit building blocks than switches (transistors)
  - Why?...

Boolean Algebra and its Relation to Digital Circuits

- To understand the benefits of "logic gates" vs. switches, we should first understand Boolean algebra
- "Traditional" algebra
  - Variable represent real numbers
  - Operators operate on variables, return real numbers
- ***Boolean Algebra***
  - Variables represent 0 or 1 only
  - Operators return 0 or 1 only
  - Basic operators
    - AND: *a AND b* returns 1 only when both a=1 and b=1
    - OR: *a OR b* returns 1 if either (or both) a=1 or b=1
    - NOT: *NOT a* returns the opposite of a (1 if a=0, 0 if a=1)

a	b	AND
0	0	0
0	1	0
1	0	0
1	1	1

a	b	OR
0	0	0
0	1	1
1	0	1
1	1	1

a	NOT
0	1
1	0

Boolean Algebra and its Relation to Digital Circuits

- Developed mid-1800's by George Boole to formalize human thought
  - Ex: "I'll go to lunch if Mary goes OR John goes, AND Sally does not go."
    - Let F represent my going to lunch (1 means I go, 0 I don't go)
    - Likewise, m for Mary going, j for John, and s for Sally
    - Then **F = (m OR j) AND NOT(s)**
  - Nice features
    - Formally evaluate
      - m=1, j=0, s=1 ➔ F = (1 OR 0) AND NOT(1) = 1 AND 0 = **0**
    - Formally transform
      - F = (m and NOT(s)) OR (j and NOT(s))
        - » Looks different, but same function
        - » We'll show transformation techniques soon

a	b	AND
0	0	0
0	1	0
1	0	0
1	1	1

a	b	OR
0	0	0
0	1	1
1	0	1
1	1	1

a	NOT
0	1
1	0

Evaluating Boolean Equations

- Evaluate the Boolean equation **F = (a AND b) OR (c AND d)** for the given values of variables a, b, c, and d:
  - Q1: a=1, b=1, c=1, d=0.
    - Answer: F = (1 AND 1) OR (1 AND 0) = 1 OR 0 = 1.
  - Q2: a=0, b=1, c=0, d=1.
    - Answer: F = (0 AND 1) OR (0 AND 1) = 0 OR 0 = 0.
  - Q3: a=1, b=1, c=1, d=1.
    - Answer: F = (1 AND 1) OR (1 AND 1) = 1 OR 1 = 1.

a	b	AND
0	0	0
0	1	0
1	0	0
1	1	1

a	b	OR
0	0	0
0	1	1
1	0	1
1	1	1

a	NOT
0	1
1	0

Converting to Boolean Equations

- Convert the following English statements to a Boolean equation
  - Q1. a is 1 and b is 1.
    - Answer: F = a AND b
  - Q2. either of a or b is 1.
    - Answer: F = a OR b
  - Q3. both a and b are not 0.
    - Answer:
      - (a) Option 1: F = NOT(a) AND NOT(b)
      - (b) Option 2: F = a OR b
  - Q4. a is 1 and b is 0.
    - Answer: F = a AND NOT(b)

Converting to Boolean Equations

- Q1. A fire sprinkler system should *spray water* if high heat is sensed **and** the system is set to enabled.
  - Answer: Let Boolean variable **h** represent "high heat is sensed," **e** represent "enabled," and F represent "spraying water." Then an equation is: **F = h AND e**.
- Q2. A car alarm should *sound* if the alarm is enabled, **and** either the car is shaken **or** the door is opened.
  - Answer: Let **a** represent "alarm is enabled," **s** represent "car is shaken," **d** represent "door is opened," and F represent "alarm sounds." Then an equation is: **F = a AND (s OR d)**.
  - **Alternatively**, assuming that our door sensor **d** represents "door is closed" instead of open (meaning d=1 when the door is closed, 0 when open), we obtain the following equation: **F = a AND (s OR NOT(d))**.

Relating Boolean Algebra to Digital Design

Boolean algebra (mid-1800s)

Boole's intent: formalize human thought

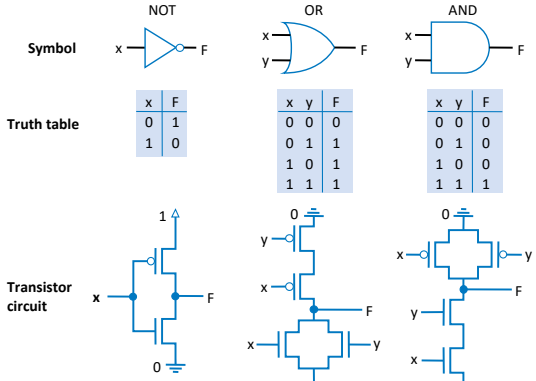
Switches (1930s)

For telephone switching and other electronic uses

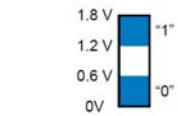
Shannon (1938)

Showed application of Boolean algebra to design of switch-based circuits

Digital design



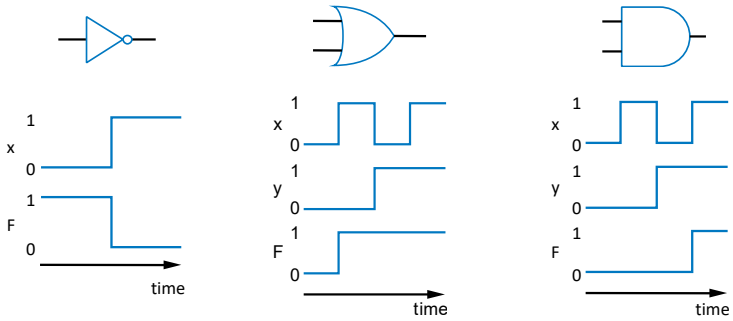
- Implement Boolean operators using transistors
  - Call those implementations ***logic gates***.
  - ***Let's us build circuits by doing math*** -- powerful concept



1 and 0 each actually corresponds to a voltage range

Note: These OR/AND implementations are inefficient; we'll show why, and show better ones later.

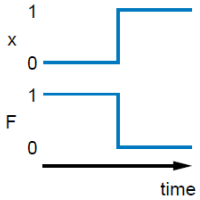
NOT/OR/AND Logic Gate Timing Diagrams

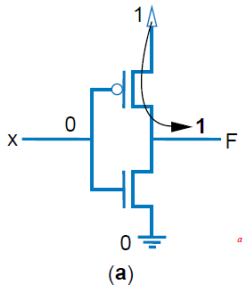


NOT Gate

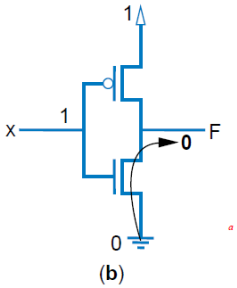


x	F
0	1
1	0



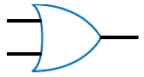


(a) When the input is 0

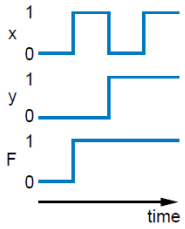


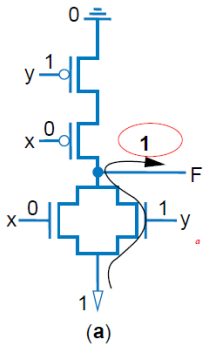
(b) When the input is 1

OR Gate

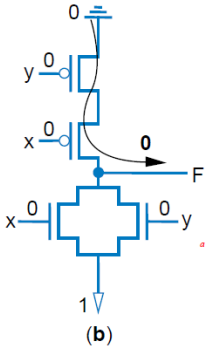


x	y	F
0	0	0
0	1	1
1	0	1
1	1	1





(a) When an input is 1

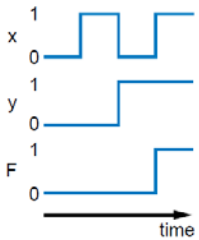


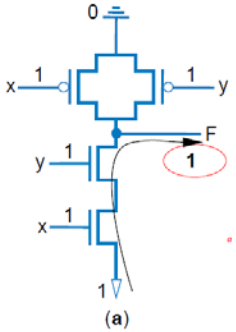
(b) When both inputs are 0

AND Gate

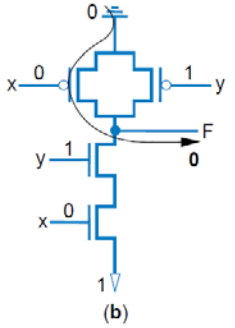


x	y	F
0	0	0
0	1	0
1	0	0
1	1	1



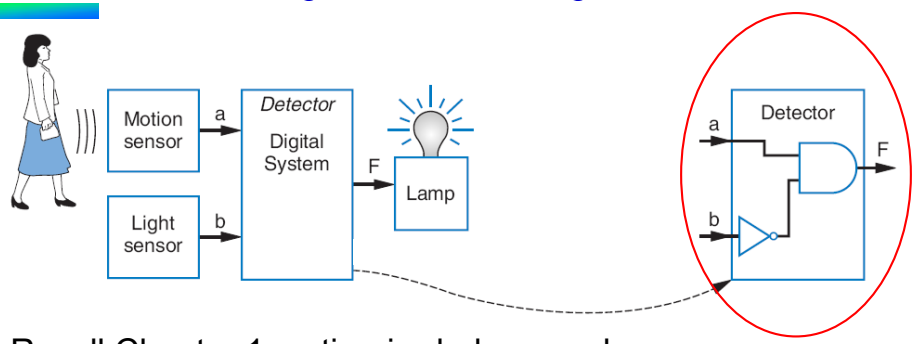


(a) When both inputs are 1



(b) When an input is 0

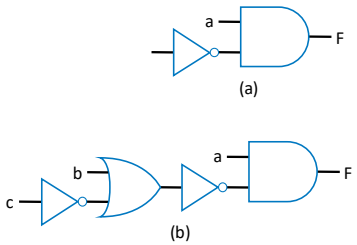
Building Circuits Using Gates



- Recall Chapter 1 motion-in-dark example
  - Turn on lamp (F=1) when motion sensed (a=1) and no light (b=0)
  - F = a AND NOT(b)
  - Build using logic gates, AND and NOT, as shown
  - We just built our first digital circuit!

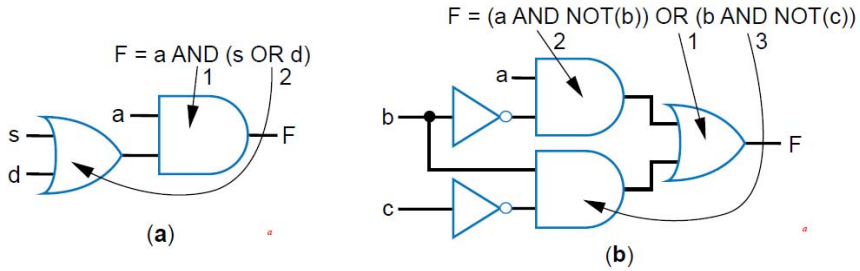
Example: Converting a Boolean Equation to a Circuit of Logic Gates

- Q: Convert the following equation to logic gates:
  F = a AND NOT(b OR NOT(c))



Start from the output, work back towards the inputs

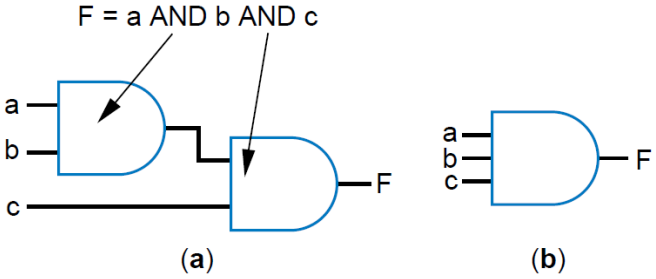
More examples



Start from the output, work back towards the inputs

Using gates with more than two inputs

- Which is better and why?



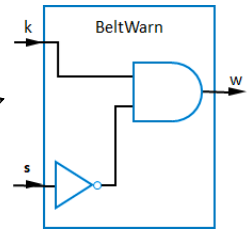
Can think of as AND(a,b,c)

Example: Seat Belt Warning Light System

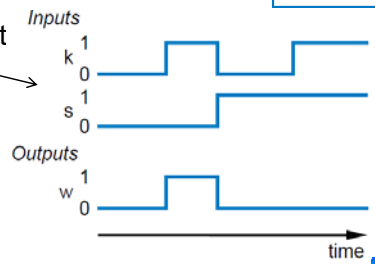


- Design circuit for warning light
- Sensors
  - s=1: seat belt fastened
  - k=1: key inserted
- **Capture** Boolean equation
  - seat belt not fastened, and key inserted

w = NOT(s) AND k



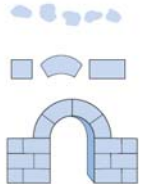
- **Convert** equation to circuit
- ***Timing diagram*** illustrates circuit behavior
  - We set inputs to any values
  - Output set according to circuit



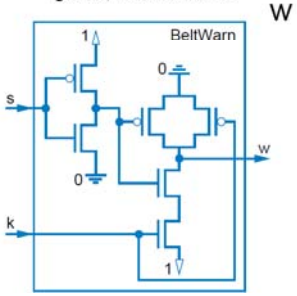
Gates vs. switches

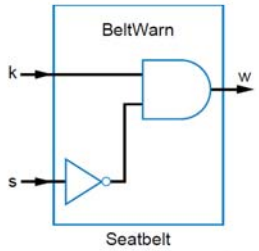
Notice

- Boolean algebra enables easy capture as equation and conversion to circuit
  - How design with switches?
  - Of course, logic gates are built from switches, but we think at level of logic gates, not switches



w = NOT(s) AND k



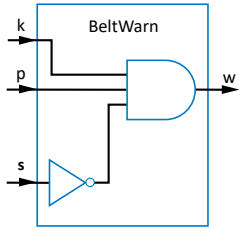


Example: Seat Belt Warning Light System

- Design circuit for warning light
- Sensors
  - s=1: seat belt fastened
  - k=1: key inserted
  - **p=1: person in seat**
- Capture Boolean equation
  - **person in seat,** and seat belt not fastened, and key inserted
- Convert equation to circuit
- Notice
  - Boolean algebra enables easy capture as equation and conversion to circuit
    - How design with switches?
    - Of course, logic gates are built from switches, but we think at level of logic gates, not switches



w = **p AND** NOT(s) AND k

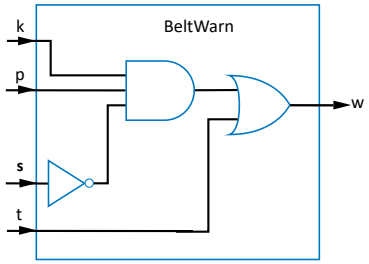


Example: Seat Belt Warning Light System

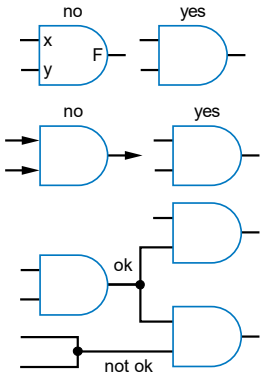
- Design circuit for warning light
- Sensors
  - s=1: seat belt fastened
  - k=1: key inserted
  - p=1: person in seat
  - **t=1: initial illumination for 5s**
- Capture Boolean equation
  - person in seat, and seat belt not fastened, and key inserted
  - Automobiles typically **initially light up all their warning lights or 5 seconds** when you first turn the key for ignition,. and 0 afterward (don't worry about who or what sets t in that way).
- Convert equation to circuit



w = **(**p AND NOT(s) AND k**) OR t**



Some Circuit Drawing Conventions



Each input and output is typically not labeled

Typically, don't draw arrows showing each direction

A single wire can be branched out

But two wires can NOT be merged into one wire.

Boolean Algebra

- By defining logic gates based on Boolean algebra, we can use **algebraic methods to manipulate circuits**
  - So let's learn some Boolean algebraic methods
- Start with notation: Writing a AND b, a OR b, and NOT(a) is cumbersome
  - Use symbols: a * b, a + b, and a' (in fact, **a * b** can be just **ab**).
    - Original: w = (p AND NOT(s) AND k) OR t
    - New: w = ps'k + t
      - Spoken as "w equals p and s prime and k, or t"
      - Or even just "w equals p s prime k, or t"
      - s' known as "complement of s"
    - While symbols come from regular algebra, *don't* say "times" or "plus"

Boolean algebra precedence, highest precedence first.

Symbol	Name	Description
( )	Parentheses	Evaluate expressions nested in parentheses first
'	NOT	Evaluate from left to right
*	AND	Evaluate from left to right
+	OR	Evaluate from left to right



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Boolean Algebra Operator Precedence

- Evaluate the following Boolean equations, assuming a=1, b=1, c=0, d=1.
  - Q1. F = a * b + c.
    - Answer: * has precedence over +, so we evaluate the equation as F = (1 *1) + 0 = (1) + 0 = 1 + 0 = 1.
  - Q2. F = ab + c.
    - Answer: the problem is identical to the previous problem, using the shorthand notation for *.
  - Q3. F = ab'.
    - Answer: we first evaluate b' because NOT has precedence over AND, resulting in F = 1 * (1') = 1 * (0) = 1 * 0 = 0.
  - Q4. F = (ac)'.
    - Answer: we first evaluate what is inside the parentheses, then we NOT the result, yielding (1*0)' = (0)' = 0' = 1.
  - Q5. F = (a + b') * c + d'.
    - Answer: Inside left parentheses: (1 + (1')) = (1 + (0)) = (1 + 0) = 1. Next, * has precedence over +, yielding (1 * 0) + 1' = (0) + 1'. The NOT has precedence over the OR, giving (0) + (1') = (0) + (0) = 0 + 0 = 0.



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Boolean Algebra Terminology

- Example equation: **F(a,b,c) = a'bc + abc' + ab + c**
- ***Variable***
  - Represents a value (0 or 1)
  - Three variables: a, b, and c
- ***Literal***
  - Appearance of a variable, in true or complemented form
  - Nine literals: a', b, c, a, b, c', a, b, and c
- ***Product term***
  - Product of literals
  - Four product terms: a'bc, abc', ab, c
- ***Sum-of-products***
  - Equation written as OR of product terms only
  - Above equation is in sum-of-products form. "F = (a+b)c + d" is not.



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Boolean Algebra Properties

- Commutative
  - a + b = b + a
  - a * b = b * a
- Distributive
  - a * (b + c) = a * b + a * c
  - a + (b * c) = (a + b) * (a + c)
    - (this one is tricky!)
- Associative
  - (a + b) + c = a + (b + c)
  - (a * b) * c = a * (b * c)
- Identity
  - 0 + a = a + 0 = a
  - 1 * a = a * 1 = a
- Complement
  - a + a' = 1
  - a * a' = 0
- To prove, just evaluate all possibilities

Example uses of the properties

- Show abc' equivalent to c'ba.
  - Use commutative property:
    - a*b*c' = a*c'*b = c'*a*b = c'*b*a = c'ba.
- Show abc + abc' = ab.
  - Use first distributive property
    - abc + abc' = ab(c+c').
  - Complement property
    - Replace c+c' by 1: ab(c+c') = ab(1).
  - Identity property
    - ab(1) = ab*1 = ab.
- Show x + x'z equivalent to x + z.
  - Second distributive property
    - Replace x+x'z by (x+x')*(x+z).
  - Complement property
    - Replace (x+x') by 1,
  - Identity property
    - replace 1*(x+z) by x+z.



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Simplify Boolean expressions using the postulates and theorems of Boolean algebra

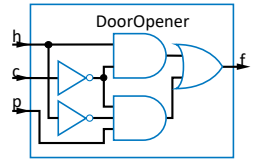
F (x, y, z)= x′y + xy′z + xyz′
= **x′y . 1** + xy′z + xyz′
= **x′y (z + z′)** + xy′z + xyz′
= **x′yz + x′yz′**+ xy′z + xyz′
= **x′yz** + xy′z + xyz′**+ x′yz′**
= **x′yz** + xy′z + yz′ (x **+ x′)**
= **x′yz** + xy′z + yz′. 1
= **x′yz** + xy′z + yz′
= **x′yz** + yz′ + xy′z
= y(**x′z** + z′) + xy′z
= y(**x′** + z′) (**z** + z′) + xy′z
= y(**x′** + z′) + xy′z
= **x′y** + yz′ + xy′z

F (x, y, z)= x′y + xy′z + xyz′
= y (x′ + xz′) + xy′z
= y **(x′ + x)** (x′ + z′) + xy′z
= y **. 1** . (x′ + z′) + xy′z
= x′y + yz′ + xy′z

Both algebraic manipulations lead to same result. Better approach?

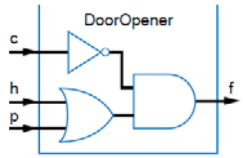
Example that Applies Boolean Algebra Properties

- Want automatic door opener circuit (e.g., for grocery store)
  - Output: f=1 opens door
  - Inputs:
    - p=1: person detected
    - h=1: switch forcing hold open
    - c=1: key forcing closed
  - Want open door when
    - h=1 and c=0, or
    - h=0 and p=1 and c=0
  - Equation: f = hc' + h'pc'



- Can the circuit be simplified?
  - f = c'hp + c'hp' + c'h'p
- Use Boolean algebra:

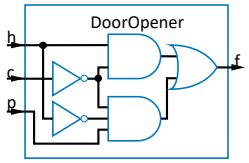
f = hc' + h'pc'
f = c'h + c'h'p (by the commutative property)
f = c'(h + h'p) (by the first distrib. property)
f = c'((h+h')*(h+p)) (2nd distrib. prop.; tricky one)
f = c'((1)*(h + p)) (by the complement property)
f = c'(h+p) (by the identity property)



Simplified circuit

Example that Applies Boolean Algebra Properties

- Want automatic door opener circuit (e.g., for grocery store)
  - Output: f=1 opens door
  - Inputs:
    - p=1: person detected
    - h=1: switch forcing hold open
    - c=1: key forcing closed
  - Want open door when
    - h=1 and c=0, or
    - h=0 and p=1 and c=0
  - Equation: f = hc' + h'pc'



- Found inexpensive chip that computes:
  - f = c'hp + c'hp' + c'h'p



  - Can we use it?
    - Is it the same as f = hc' + h'pc'?
- Use Boolean algebra:

f = c'hp + c'hp' + c'h'p
f = c'h(p + p') + c'h'p (by the distributive property)
f = c'h(1) + c'h'p (by the complement property)
f = c'h + c'h'p (by the identity property)
f = hc' + h'pc' (by the commutative property)
Same!

Boolean Algebra: Additional Properties

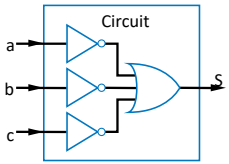
- Null elements
  - a + 1 = 1
  - a * 0 = 0
- Idempotent Law
  - a + a = a
  - a * a = a
- Involution Law
  - (a')' = a
- DeMorgan's Law
  - (a + b)' = a'b'
  - (ab)' = a' + b'
  - Very useful!

To prove, just evaluate all possibilities

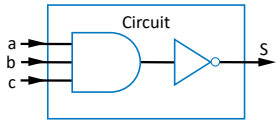
Example Applying DeMorgan's Law
Aircraft lavatory sign example

- Behavior
  - Three lavatories, each with sensor (a, b, c), equals 1 if door locked
  - Light **"Available"** sign (S) if any lavatory available
- Equation and circuit
  - S = a' + b' + c'
- Transform
  - (abc)' = a'+b'+c' (by DeMorgan's Law)
  - S = (abc)'
- New equation and circuit
- Alternative: Instead of lighting "Available," light **"Occupied"**
  - Opposite of "Available" function S = a' + b' + c'
  - So S' = (a' + b' + c')'
    - S' = (a')' * (b')' * (c')' (by DeMorgan's Law)
    - S' = a * b * c (by Involution Law)
  - Makes intuitive sense
    - Occupied if all doors are locked





Aircraft lavatory sign circuit.



Circuit after applying DeMorgan's Law

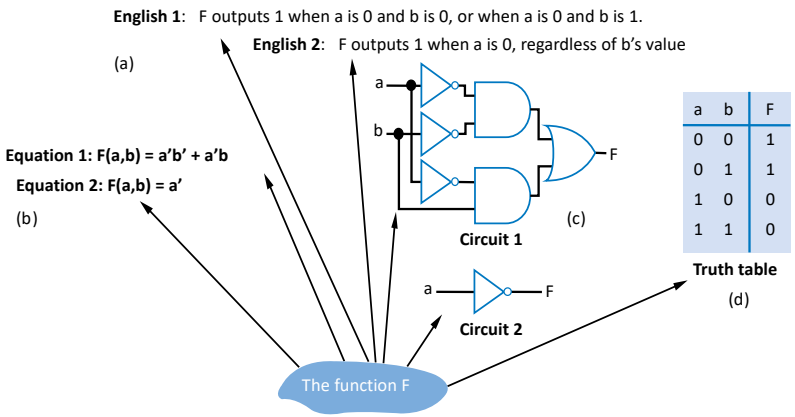
Example Applying Properties

- For door opener f = c'(h+p) , *prove* door stays closed (f=0) when c=1
  - f = c'(h+p)
  - *Let c = 1* (door forced closed)
  - f = 1'(h+p)
  - f = 0(h+p)
  - f = 0h + 0p (by the distributive property)
  - f = 0 + 0 (by the null elements property)
  - f = 0

Complement of a Function

- Commonly want to find complement (inverse) of function F
  - 0 when F is 1; 1 when F is 0
- Use DeMorgan's Law repeatedly
  - Note: DeMorgan's Law defined for more than two variables, e.g.:
    - (a + b + c)' = (abc)'
    - (abc)' = (a' + b' + c')
- Complement of f = w'xy + wx'y'z'
  - f ' = (w'xy + wx'y'z')'
  - f ' = (w'xy)'(wx'y'z')' (by DeMorgan's Law)
  - f ' = (w+x'+y')(w'+x+y+z) (by DeMorgan's Law)
- Can then expand into sum-of-products form

Representations of Boolean Functions



- A function can be represented in different ways
  - Above shows seven representations of the same functions F(a,b), using four different methods: **English**, **Equation**, **Circuit**, and **Truth Table**

Truth Table Representation of Boolean Functions

- Define value of F for each possible combination of input values
  - 2-input function: 4 rows
  - 3-input function: 8 rows
  - 4-input function: 16 rows
- Q: Use truth table to define function F(a,b,c) that is 1 when abc is 5 or greater in binary

a	b	F
0	0	
0	1	
1	0	
1	1	

(a)

a	b	c	F
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

(b)

a	b	c	d	F
0	0	0	0	
0	0	0	1	
0	0	1	0	
0	0	1	1	
0	1	0	0	
0	1	0	1	
0	1	1	0	
0	1	1	1	
1	0	0	0	
1	0	0	1	
1	0	1	0	
1	0	1	1	
1	1	0	0	
1	1	0	1	
1	1	1	0	
1	1	1	1	

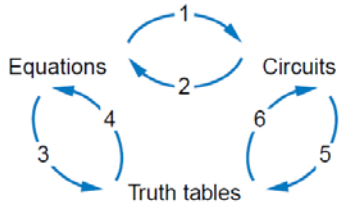
(c)

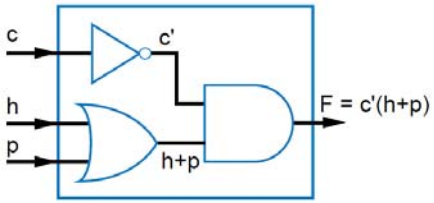
a	b	c	F
0	0	0	**0**
0	0	1	**0**
0	1	0	**0**
0	1	1	**0**
1	0	0	**0**
1	0	1	**1**
1	1	0	**1**
1	1	1	**1**

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Converting among Representations

- Can convert from any representation to another
- Common conversions
  - Equation to circuit (we did this earlier)
  - Circuit to equation
    - Start at inputs, write expression of each gate output





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Converting among Representations

- Can convert from any representation to any other
- Common conversions
  - Equation to circuit (we did this earlier)
  - Truth table to equation (which we can convert to circuit)
    - Easy -- just OR each input term that should output 1
  - Equation to truth table
    - Easy -- just evaluate equation for each input combination (row)
    - Creating intermediate columns helps

Q: Convert to truth table: F = a'b' + a'b

Inputs				Output
a	b	a' b'	a' b	F
0	0	1	0	**1**
0	1	0	1	**1**
1	0	0	0	**0**
1	1	0	0	**0**

Inputs		Outputs	Term
a	b	F	F = sum of
0	0	1	a'b'
0	1	1	a'b
1	0	0	
1	1	0	

F = a'b' + a'b

Q: Convert to equation

a	b	c	F	
0	0	0	0	
0	0	1	0	
0	1	0	0	
0	1	1	0	
1	0	0	0	
1	0	1	1	ab'c
1	1	0	1	abc'
1	1	1	1	abc

F = ab'c + abc' + abc

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Standard Representation: Truth Table

- How can we determine if two functions are the same?
  - Recall automatic door example
    - Same as f = hc' + h'pc'?
    - Used algebraic methods
    - But if we failed, does that prove *not* equal? No.
- Solution: Convert to truth tables
  - Only ONE truth table representation of a given function
    - ***Standard*** representation -- for given function, only one version in standard form exists

f = c'hp + c'hp' + c'h'

f = c'h(p + p') + c'h'p

f = c'h(1) + c'h'p

f = c'h + c'h'p

(what if we stopped here?)

f = hc' + h'pc'

Q: Determine if F=ab+a' is same function as F=a'b'+a'b+ab, by converting each to truth table first

F = ab + a'		
a	b	F
0	0	1
0	1	1
1	0	0
1	1	1

Same

F = a'b' + a'b + ab		
a	b	F
0	0	1
0	1	1
1	0	0
1	1	1

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Canonical Form -- Sum of Minterms

- Truth tables too big for numerous inputs
- Use standard form of equation instead
  - Known as ***canonical form***
  - Regular algebra: group terms of polynomial by power
    - $ax^2 + bx + c$ $(3x^2 + 4x + 2x^2 + 3 + 1 \rightarrow 5x^2 + 4x + 4)$
  - Boolean algebra: create sum of minterms
    - ***Minterm***: product term with every function literal appearing exactly once, in true or complemented form
    - Just multiply-out equation until sum of product terms
    - Then expand each term until all terms are minterms

Q: Determine if F(a,b)=ab+a' is same function as F(a,b)=a'b'+a'b+ab, by converting first equation to canonical form (second already in canonical form)

F = ab+a' (already sum of products)
F = ab + a'(b+b') (expanding term)
F = ab + a'b + a'b' (SAME -- same three terms as other equation)

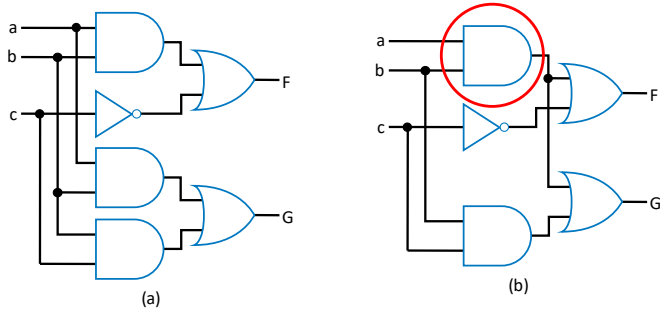
Compact Sum of Minterms Representation

- List each minterm as a number
- Number determined from the binary representation of its variables' values
  - a'bcde corresponds to 01111, or 15
  - abcde' corresponds to 11110, or 30
  - abcde corresponds to 11111, or 31

- Thus, H = a'bcde + abcde' + abcde can be written as:
  - H = Σm(15,30,31)
  - "H is the sum of minterms 15, 30, and 31"

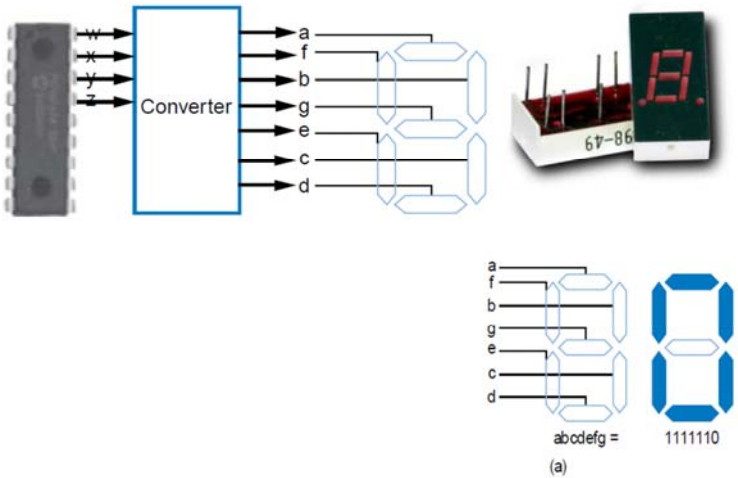
Multiple-Output Circuits

- Many circuits have more than one output
- Can give each a separate circuit, or can share gates
- Ex: F = **ab** + c', G = **ab** + bc



Option 1: Separate circuits Option 2: Shared gates

Multiple-Output Example: BCD to 7-Segment Converter



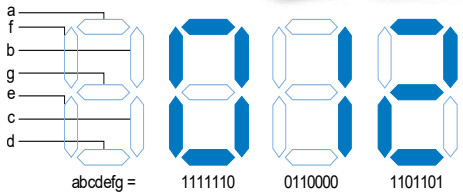
Multiple-Output Example:
BCD to 7-Segment Converter



TABLE 2-4 4-bit binary number to seven-segment display truth table

w	x	y	z	a	b	c	d	e	f	g
0	0	0	0	1	1	1	1	1	1	0
0	0	0	1	0	1	1	0	0	0	0
0	0	1	0	1	1	0	1	1	0	1
0	0	1	1	1	1	1	1	0	0	1
0	1	0	0	0	1	1	0	0	1	1
0	1	0	1	1	0	1	1	0	1	1
0	1	1	0	1	0	1	1	1	1	1
0	1	1	1	1	1	1	0	0	0	0
1	0	0	0	1	1	1	1	1	1	1
1	0	0	1	1	1	1	1	0	1	1
1	0	1	0	0	0	0	0	0	0	0
1	0	1	1	0	0	0	0	0	0	0
1	1	0	0	0	0	0	0	0	0	0
1	1	0	1	0	0	0	0	0	0	0
1	1	1	0	0	0	0	0	0	0	0
1	1	1	1	0	0	0	0	0	0	0





a = w'x'y'z' + w'x'yz' + w'x'yz + w'xy'z + w'xyz' + w'xyz + wx'y'z' + wx'y'z

b = w'x'y'z' + w'x'y'z + w'x'yz' + w'x'yz + w'xy'z' + w'xyz + wx'y'z' + wx'y'z

- Minimize the logic for each equation before creating each of the circuits
- Equations for a and b have several terms in common. The wx'y'z' is common to all.

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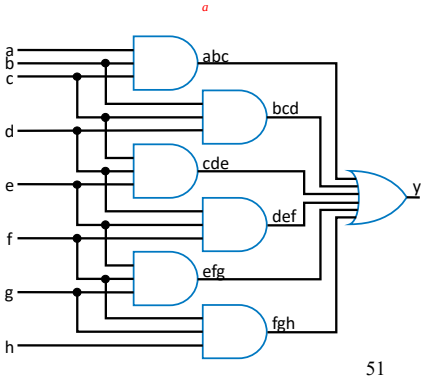
Combinational Logic Design Process

Step		Description
Step 1 Capture Behavior	**Capture** the function	Create a truth table or equations, ***whichever is most natural for the given problem***, to describe the desired behavior of the combinational logic.
Step 2 Convert To Circuit	2A: **Create** equations	This substep is only necessary if you captured the function using a truth table instead of equations. Create an equation for each output by ORing all the minterms for that output. Simplify the equations if desired.
	2B: **Implement** as a gate-based circuit	For each output, create a circuit corresponding to the output's equation. (Sharing gates among multiple outputs is OK optionally.)

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Example: Three 1s Detector

- Problem: Detect three consecutive 1s in 8-bit input: abcdefgh
  - 000**111**01 → 1 10101011 → 0
    11110000 → 1
  - **Step 1: Capture** the function
    - Truth table or equation?
      - Truth table too big: 2^8=256 rows
      - Equation: create terms for each possible case of three consecutive 1s
    - **y = abc + bcd + cde + def + efg + fgh**
  - **Step 2: Convert** to equation -- already done
  - **Step 3: Implement** as a gate-based circuit

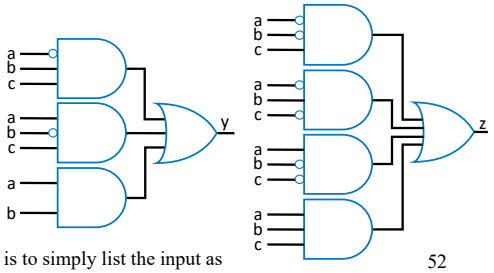


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Example: Number of 1s Count

- Problem: Output in binary on two outputs yz the number of 1s on three inputs
  - 010 → 01 101 → 10 000 → 00
  - **Step 1: Capture** the function
    - Truth table or equation?
      - Truth table is straightforward
  - **Step 2: Convert** to equation
    - y = a'bc + ab'c + abc' + abc
    - z = a'b'c + a'bc' + ab'c' + abc
  - **Step 3: Implement** as a gate-based circuit

Inputs			(# of 1s)	Outputs	
a	b	c		y	z
0	0	0	(0)	0	0
0	0	1	(1)	0	1
0	1	0	(1)	0	1
0	1	1	(2)	1	0
1	0	0	(1)	0	1
1	0	1	(2)	1	0
1	1	0	(2)	1	0
1	1	1	(3)	1	1

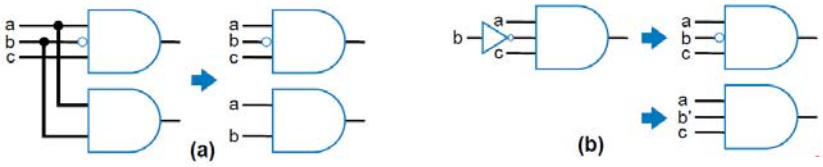


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An alternative simplification is to simply list the input as complemented, like b' instead of b and the inversion bubble.

Simplifying Notations

- Used in previous circuit

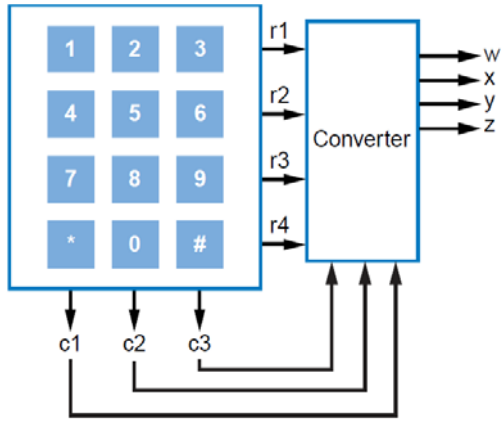


List inputs multiple times
→Less wiring in drawing

Draw inversion bubble rather than inverter. Or list input as complemented.

12-button keypad to 4-bit code converter

- Truth table
  - Number of row?
  - Number of columns?



12-button keypad to 4-bit code converter

TABLE 2.7 Informal table for the 12-button keypad to 4-bit code converter.

Button	Signals		4-bit code outputs			
			w	x	y	z
1	r1	c1	0	0	0	1
2	r1	c2	0	0	1	0
3	r1	c3	0	0	1	1
4	r2	c1	0	1	0	0
5	r2	c2	0	1	0	1
6	r2	c3	0	1	1	0
7	r3	c1	0	1	1	1
8	r3	c2	1	0	0	0
9	r3	c3	1	0	0	1
*	r4	c1	1	0	1	0
0	r4	c2	0	0	0	0
#	r4	c3	1	0	1	1
(none)			1	1	1	1

w = r3c2 + r3c3 + r4cl + r4c3 + rl'r2'r3'r4'cl' c2'c3'

x = r2cl + r2c2 + r2c3 + r3cl + rl' r2'r3'r4'c1'c2'c3'

y = rlc2 + rlc3 + r2c3 + r3cl + r4cl + r4c3 + rl'r2'r3'r4'cl'c2'c3'

z = rlcl + rlc3 + r2c2 + r3cl + r3c3 + r4c3 + rl'r2'r3'r4'cl'c2'c3'

Sprinkler valve controller

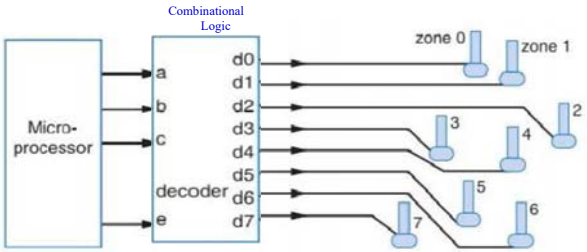
- Supports up to 8 zones.
- Microprocessor
  - only has 4 output pins available to control the valves
  - use 1 pin to indicate whether a valve should be opened
  - use 3 other pins to output the active zone (0, 1, ... , 7) in binary

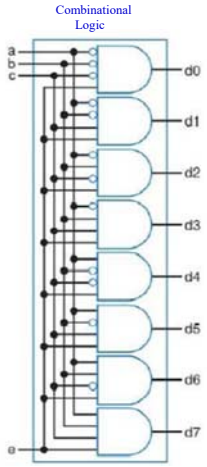
Sprinkler valve controller

- Step 1: Capture behavior

$d_0 = a'b'c'e$
$d_1 = a'b'ce$
$d_2 = a'bc'e$
$d_3 = a'bce$
$d_4 = ab'c'e$
$d_5 = ab'ce$
$d_6 = abc'e$
$d_7 = abce$

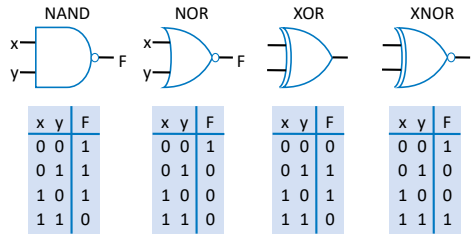
- Step 2A: Create equations
- Step 2B: Implement as a gate· based circuit





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More Gates

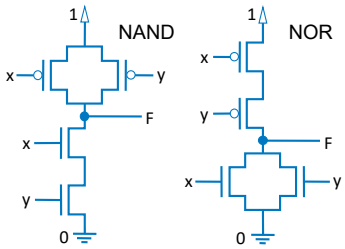


x	y	F
0	0	1
0	1	1
1	0	1
1	1	0

x	y	F
0	0	1
0	1	0
1	0	0
1	1	0

x	y	F
0	0	0
0	1	1
1	0	1
1	1	0

x	y	F
0	0	1
0	1	0
1	0	0
1	1	1

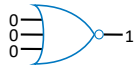


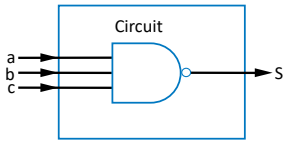
- NAND: Opposite of AND ("NOT AND")
- NOR: Opposite of OR ("NOT OR")
- XOR: Exactly 1 input is 1, for 2-input XOR. (For more inputs -- odd number of 1s)
- XNOR: Opposite of XOR ("NOT XOR")
- NAND same as AND with power & ground switched
  - Why? nMOS conducts 0s well, but not 1s (reasons beyond our scope) -- so NAND more efficient
- Likewise, NOR same as OR with power/ground switched
- AND in CMOS: NAND with NOT
- OR in CMOS: NOR with NOT
- So, NAND/NOR implementations are more common

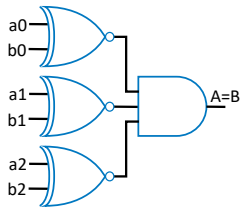
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More Gates: Example Uses

- Aircraft lavatory sign example
  - S = (abc)'
- Detecting all 0s
  - Use NOR
- Detecting equality
  - Use XNOR
- Detecting odd # of 1s
  - Use XOR
  - Useful for generating "parity" bit common for detecting errors







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Completeness of NAND

- Any Boolean function can be implemented *using just NAND gates.* Why?
  - Need AND, OR, and NOT
  - NOT: 1-input NAND (or 2-input NAND with inputs tied together)
  - AND: NAND followed by NOT
  - OR: NAND preceded by NOTs



- Likewise, for NOR

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Number of Possible Boolean Functions

- How many possible functions of 2 variables?
  - 2^2 rows in truth table, 2 choices for each
  - $2^{(2^2)} = 2^4 = 16$ possible functions
- N variables
  - 2^N rows
  - $2^{(2^N)}$ possible functions

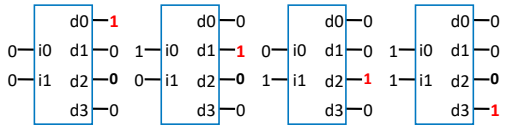
a	b	F
0	0	0 or 1 2 choices
0	1	0 or 1 2 choices
1	0	0 or 1 2 choices
1	1	0 or 1 2 choices

$2^4 = 16$ possible functions

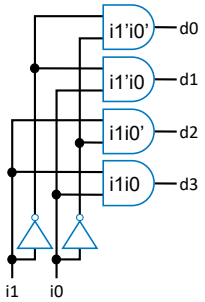
a	b	f0	f1	f2	f3	f4	f5	f6	f7	f8	f9	f10	f11	f12	f13	f14	f15
0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1
0	1	0	0	0	0	1	1	1	1	0	0	0	0	1	1	1	1
1	0	0	0	1	1	0	0	1	1	0	0	1	1	0	0	1	1
1	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1
		0	a AND b		a		b	a XOR b	a OR b	a NOR b	a XNOR b	b'		a'		a NAND b	1

Decoders

- **Decoder**: Popular combinational logic building block, in addition to logic gates
  - Converts input binary number to one high output
- 2-input decoder: four possible input binary numbers
  - So has four outputs, one for each possible input binary number



- Internal design
  - AND gate for each output to detect input combination
- n-input decoder: 2^n outputs



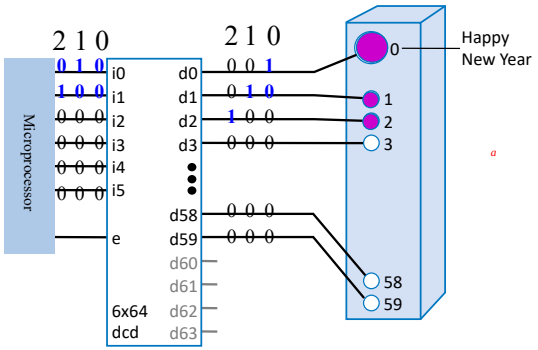
Decoders with Enable

- Decoder with enable **e**
  - Outputs all 0 if e=0
  - Regular behavior if e=1



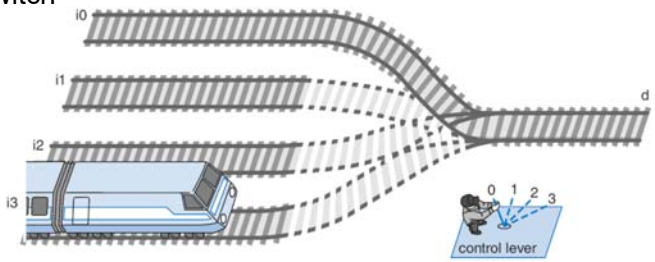
Decoder Example

- New Year's Eve Countdown Display
  - Microprocessor counts from 59 down to 0 in binary on 6-bit output
  - Want illuminate one of 60 lights for each binary number
  - Use 6x64 decoder
    - 4 outputs unused

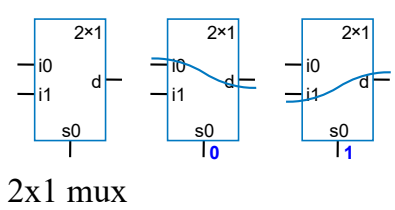


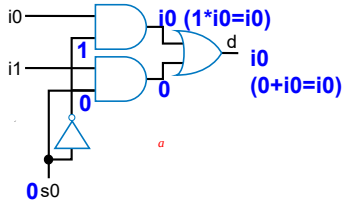
Multiplexor (Mux)

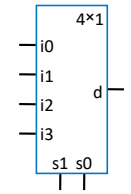
- Mux: Another popular combinational building block
  - Routes one of its N data inputs to its one output, based on binary value of select inputs
    - 4 input mux → needs 2 select inputs to indicate which input to route through
    - 8 input mux → 3 select inputs
    - N inputs → $\log_2(N)$ selects
  - Like a railyard switch

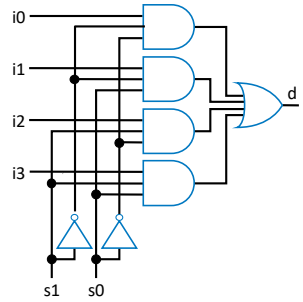


Mux Internal Design





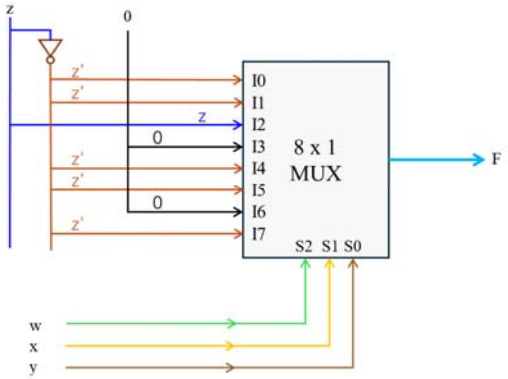




Implement the Boolean function with an 8 ×1 MUX

- F(w, x, y, z) = ∑ (0, 2, 5, 8, 10, 14)

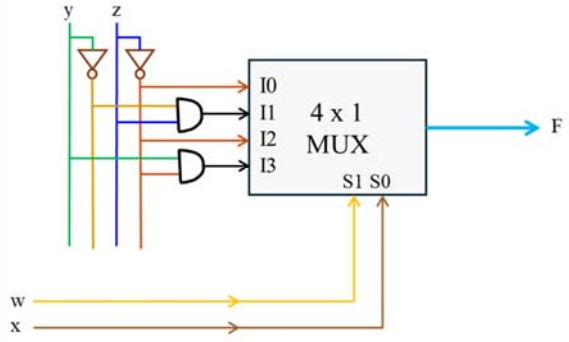
m_i	S_2 w	S_1 x	S_0 y	z	F	F_S(z)	I_S
m_0	0	0	0	0	1	z′	I_0
m_1	0	0	0	1	0		
m_2	0	0	1	0	1	z′	I_1
m_3	0	0	1	1	0		
m_4	0	1	0	0	0	z	I_2
m_5	0	1	0	1	1		
m_6	0	1	1	0	0	0	I_3
m_7	0	1	1	1	0		
m_8	1	0	0	0	1	z′	I_4
m_9	1	0	0	1	0		
m_{10}	1	0	1	0	1	z′	I_5
m_{11}	1	0	1	1	0		
m_{12}	1	1	0	0	0	0	I_6
m_{13}	1	1	0	1	0		
m_{14}	1	1	1	0	1	z′	I_7
m_{15}	1	1	1	1	0		



Implement the Boolean function with a 4 ×1 MUX

- F(w, x, y, z) = ∑ (0, 2, 5, 8, 10, 14)

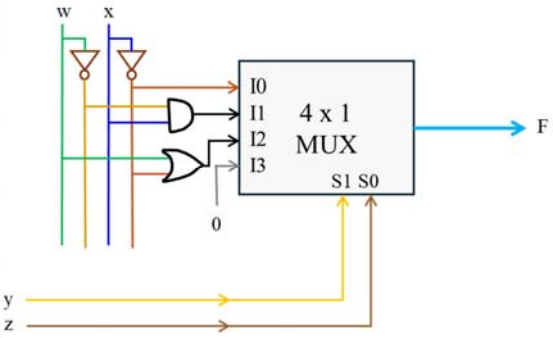
m_i	S_1 w	S_0 x	y	z	F	F_S(y, z)	I_S
m_0	0	0	0	0	1	$F_0 = z'$	I_0
m_1	0	0	0	1	0		
m_2	0	0	1	0	1		
m_3	0	0	1	1	0		
m_4	0	1	0	0	0	$F_1 = y'z$	I_1
m_5	0	1	0	1	1		
m_6	0	1	1	0	0		
m_7	0	1	1	1	0		
m_8	1	0	0	0	1	$F_2 = z'$	I_2
m_9	1	0	0	1	0		
m_{10}	1	0	1	0	1		
m_{11}	1	0	1	1	0		
m_{12}	1	1	0	0	0	$F_3 = yz'$	I_3
m_{13}	1	1	0	1	0		
m_{14}	1	1	1	0	1		
m_{15}	1	1	1	1	0		



Implement the Boolean function with a 4 ×1 MUX

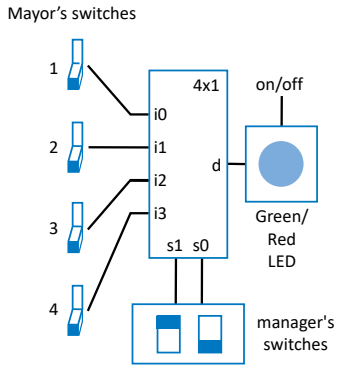
- F(w, x, y, z) = ∑ (0, 2, 5, 8, 10, 14)

m_i	S_1 y	S_0 z	w	x	F	$F_S(y, z)$	I_S
m_0	0	0	0	0	1	$F_0 = x'$	I_0
m_4	0	0	0	1	0		
m_8	0	0	1	0	1		
m_{12}	0	0	1	1	0		
m_1	0	1	0	0	0	$F_1 = w'x$	I_1
m_5	0	1	0	1	1		
m_9	0	1	1	0	0		
m_{13}	0	1	1	1	0		
m_2	1	0	0	0	1	$F_2 = w+x'$	I_2
m_6	1	0	0	1	0		
m_{10}	1	0	1	0	1		
m_{14}	1	0	1	1	1		
m_3	1	1	0	0	0	$F_3 = 0$	I_3
m_7	1	1	0	1	0		
m_{11}	1	1	1	0	0		
m_{15}	1	1	1	1	0		

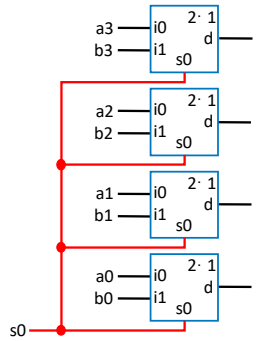


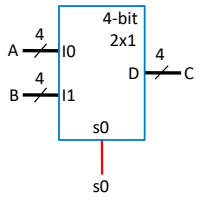
Mux Example

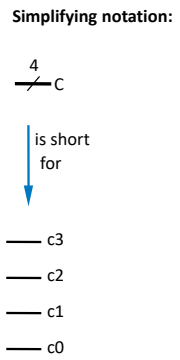
- City mayor can set four switches up or down, representing his/her vote on each of four proposals, numbered 0, 1, 2, 3
- City manager can display any such vote on large green/red LED (light) by setting two switches to represent binary 0, 1, 2, or 3
- Use 4x1 mux



Muxes Commonly Together -- N-bit Mux

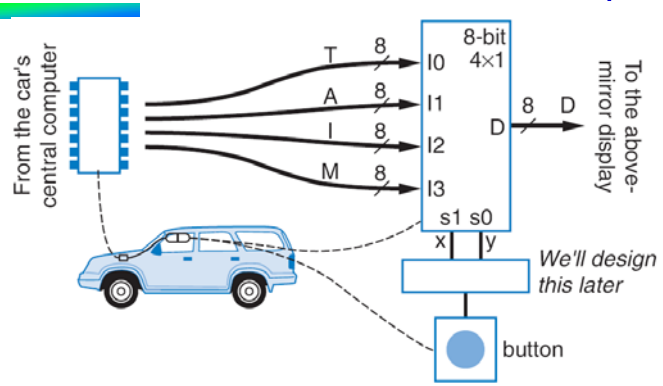






- Ex: Two 4-bit inputs, A (a3 a2 a1 a0), and B (b3 b2 b1 b0)
  - 4-bit 2x1 mux (just four 2x1 muxes sharing a select line) can select between A or B

N-bit Mux Example

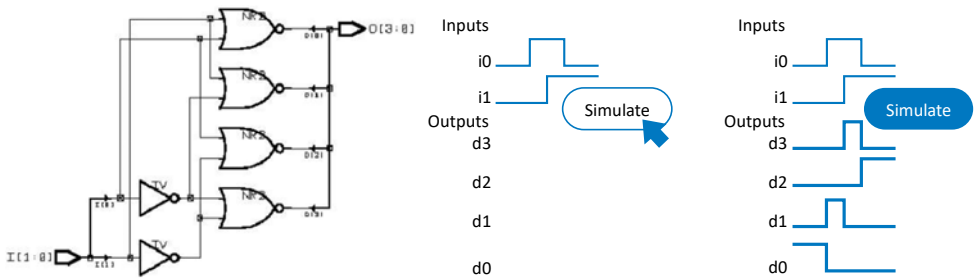




- Four possible display items
  - Temperature (T), Average miles-per-gallon (A), Instantaneous mpg (I), and Miles remaining (M) -- each is 8-bits wide
  - Choose which to display using two inputs x and y
  - Use **8-bit 4x1 mux**

Additional Considerations

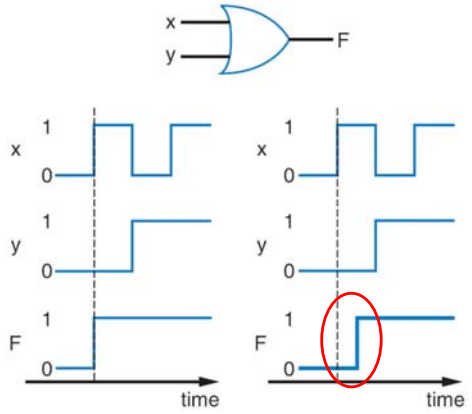
Schematic Capture and Simulation



- **Schematic capture**
  - Computer tool for user to capture logic circuit graphically
- **Simulator**
  - Computer tool to show what circuit outputs would be for given inputs
    - Outputs commonly displayed as **waveform**

Additional Considerations

Non-Ideal Gate Behavior – Delay



- Real gates have some delay
  - Outputs don't change immediately after inputs change

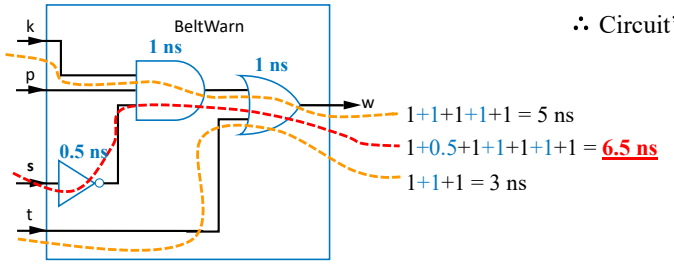
Additional Considerations

Finding the Circuit's Deay – Critical Path

- The delay of a circuit, known as ***circuit delay***, is the delay of the longest path from input to output, known as the circuit's ***critical path***.

∵ Critical Path Delay = 6.5 ns

∴ Circuit's Delay = 6.5 ns



Assume that each wire segment also introduces a delay of 1 ns in path delay, in addition delays of the path gates.

Additional Considerations

Active High and Active Low Inputs

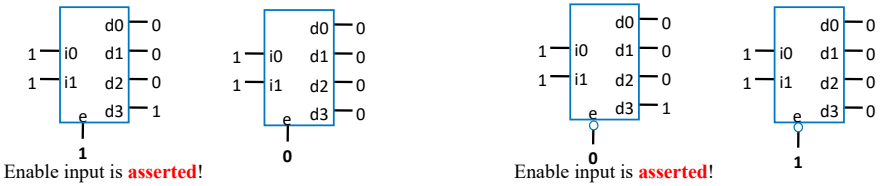
- Component inputs can generally be divided into two types.
  - ***Control inputs*** influence the behavior of the component, such as the two select inputs of a 4x1 mux, or the enable input of a 3x8 decoder.
  - ***Data inputs*** flow through the component, such as the 4 data inputs of a 4x1 mux, the 3 data inputs of a 3x8 decoder, or the inputs of any logic gate.

- Some control inputs involve the notion of being *active*
  - *when* the input is at one of its two levels, the input is carrying out its purpose i.e., the input is **active**
  - At the other level, the input is **inactive**

Additional Considerations

Active High and Active Low Inputs – Decoder

- Decoder with active-high enable input **e**
  - Outputs all 0 if e=0
  - Regular behavior if e=1



Enable input is **asserted**!

Decoder with active high enable input:
(a) e=0 : disabled, all outputs 0,
(b) e=1: enabled, normal output.

- Decoder with active-low enable input **e**
  - Outputs all 0 if e=1
  - Regular behavior if e=0

Enable input is **asserted**!

Decoder with active low enable input:
(a) e=1 : disabled, all outputs 0,
(b) e=0: enabled, normal output.

- Use the term *assert* with control input to mean setting a control input to the value that activates the associated operation.
- Using the term assert avoids possible confusion that could occur when some control inputs are active-high and others are active-low.



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Chapter Summary

- Combinational circuits
  - Circuit whose outputs are function of present inputs
    - No "state"
- Switches: Basic component in digital circuits
- Boolean logic gates: AND, OR, NOT -- Better building block than switches
  - Enables use of Boolean algebra to design circuits
- Boolean algebra: uses true/false variables/operators
- Representations of Boolean functions: Can translate among
- Combinational design process: Translate from equation (or table) to circuit through well-defined steps
- More gates: NAND, NOR, XOR, XNOR also useful
- Muxes and decoders: Additional useful combinational building blocks



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